VEHICLE SECURITY AND AUTHENTICATION SYSTEM AND METHOD

FIELD OF THE INVENTION

[0001]
The present invention relates generally to automotive cybersecurity, vehicle access control, electronic immobilizers, and multi-factor authentication systems. More specifically, the invention relates to an out-of-band physical-presence verification method utilizing near-ultrasonic acoustic steganography transmitted within an enclosed passenger cabin to gate electronic control unit (ECU) starter circuits and high-voltage interlocks, mitigating wireless radio frequency (RF) keyless relay station attacks and physical key fob theft without requiring secondary radio frequency hardware.

DEFINITIONS AND TERMINOLOGY

[0002]
For the purposes of this specification and the appended claims, the following terms are defined:

[0003]
"Near-Ultrasonic Band": Acoustic frequencies situated between approximately 16.0 kHz and 22.0 kHz that reside near or above the nominal threshold of adult human hearing while remaining within the reproduction and capture capabilities of standard automotive audio equipment operating at 44.1 kHz or 48.0 kHz sampling rates.

[0004] "Acoustic Boundary Attenuation": The structural transmission loss, absorption, and spatial confinement of acoustic energy achieved by an enclosed vehicle cabin structure 100 (comprising laminated safety glass, steel and aluminum body panels, elastomeric door weatherstripping, and acoustic headliners), which attenuates near-ultrasonic transmissions escaping the vehicle cabin below an operable signal-to-noise ratio (SNR) for external receivers, rendering external acoustic capture insufficient to reconstruct de-modulable bitstreams in ambient environments.

[0005]
"Spectral Guard Band": An unmodulated frequency buffer band 220 of at least 1.5 kHz (preferably 1.5 kHz to 3.0 kHz wide) established between the highest cutoff frequency of a low-pass-filtered audible carrier 210 (e.g., <= 14.0 kHz) and the

lowest frequency carrier of an ultrasonic data payload 230 (e.g., >= 16.0 kHz) i.e. (e.g., 16.0 kHz or greater).

[0006]
"State-Gated Dormant Fallback": A secondary authentication architecture wherein manual override interfaces (such as touchscreens, center-console interfaces, or steering-wheel buttons) remain fully unpowered, unclocked, or software-isolated until an explicit timeout of a primary acoustic authentication window occurs, preventing parallel brute-force attack vectors.

[0007]
"Physical-Presence Listening State": A gated operating state (Stage 1, 410) of an electronic control unit (ECU 130) wherein high-voltage propulsion contactors or engine starter motor relays 140 remain disabled while one or more interior cabin microphones 120 are actively energized for an ingress listening window (e.g., configurable between 10.0 and 60.0 seconds, and preferably between 30.0 and 45.0 seconds) to capture an in-cabin acoustic token.

[0008]
"Bidirectional Acoustic Challenge-Response": A protocol exchange wherein a vehicle interior transducer 125, 602 emits an acoustic challenge seed into the cabin, and a user device 110, 606 must capture said seed inside the cabin to compute and return an acoustic response token within a predetermined round-trip latency window (Delta-t <= 1.5 seconds) i.e. (delta-t less than or equal to 1.5 seconds).

[0009]
"Two-Factor Authentication (2FA) Ignition Gating": A security architecture requiring the concurrent presence and cryptographic validation of two distinct, independent credentials, a primary wireless ingress credential (e.g., physical RF key fob or BLE digital key 150) and a secondary physical-presence acoustic token—before enabling vehicle propulsion circuits 140.

BACKGROUND OF THE INVENTION AND PRIOR ART

3.1 The Technical Problem: Keyless RF Relay Attacks and Physical Key Compromise

[0010]
Modern motor vehicles commonly incorporate Passive Keyless Entry and Start (PKES) systems, allowing an operator to unlock and start a vehicle while retaining a radio-frequency (RF) key fob or mobile digital key in a pocket, bag, or wearable.

Standard PKES architectures communicate across wireless RF frequencies, typically ranging between 315 MHz, 433 MHz, and 2.4 GHz (such as Bluetooth Low Energy).

[0011]
These conventional systems suffer from two primary physical-layer security vulnerabilities:

[0012] Wireless Relay Station Attacks (RSA): In a typical relay attack, an unauthorized actor 160 positions an RF amplifier and transceiver in proximity to a legitimate key fob 150 (e.g., outside the exterior wall of an owner's residence) and establishes a wireless bridge to a secondary transceiver held by an accomplice adjacent to the target vehicle. Because RF signals penetrate domestic walls, glass windows, and automotive body panels, the vehicle's Electronic Control Unit (ECU 130) or Body Control Module (BCM) falsely determines that the physical key fob is in direct proximity, consequently unlocking doors and authorizing engine ignition.

[0013]
Physical Key Fob Theft and Infiltration: In residential burglaries, letterbox intrusions, or locker thefts, an authentic physical key fob 150 is physically stolen by an unauthorized party. In conventional single-factor vehicles, physical possession of the key fob grants immediate, unrestricted engine starting and vehicle drive-away. Existing systems lack an independent, in-cabin secondary authentication layer capable of preventing unauthorized vehicle theft when the primary physical key has been physically stolen or compromised.

3.2 Shortcomings of Existing Countermeasures

[0014]
Ultra-Wideband (UWB) Time-of-Flight Radar: Modern countermeasures often implement Ultra-Wideband (IEEE 802.15.4z) transceiver chips to calculate wireless time-of-flight (ToF) distance. While effective against relay attacks, UWB introduces significant Bill-of-Materials (BOM) hardware costs, requiring specialized transceiver antennas distributed across vehicle body panels, specialized transceivers inside the key fob, and compatible mobile devices. Furthermore, UWB fails to protect against physical key theft, as a stolen UWB key will still calculate a legitimate time-of-flight distance inside the car.

[0015]
Manual PIN-to-Drive Interfaces: Secondary authentication mechanisms requiring user interaction with in-cabin touchscreens, steering-wheel button combinations, or manual keypad entries introduce substantial cognitive friction, disrupt the seamless vehicle startup experience, and leave visible latent fingerprint residues on glass displays that can be visually compromised.

3.3 Differentiation from Automotive Time-of-Flight Prior Art (Volvo Patent Family: US 10,759,383 B2, US 11,528,612 B2, and US 11,974,133 B2)

[0016]
US 10,759,383 B2 and its continuation family describe systems requiring an RF transceiver in combination with acoustic time-of-flight (ToF) distance-ranging and echo reflection measurements using exterior vehicle transducers (such as ultrasonic parking sensors) to authorize door-unlock requests. In contrast to systems relying upon acoustic time-of-flight distance ranging, echo reflections, or exterior transducers, the present invention operates independently of acoustic distance calculation and requires no exterior acoustic sensors. The present invention relies upon structural acoustic boundary attenuation and spatial containment within an enclosed vehicle cabin 100 (glass, sheet metal, acoustic headliner, and rubber weatherstripping) to prevent external relaying, operating via interior cabin microphones 120 to gate ECU starter circuits 140.

3.4 Differentiation from Academic Sound-Proximity Prior Art (Choi et al., 2018)

[0017]
Choi, Seo, and Lee (Journal of Advanced Transportation, 2018, "Sound-Proximity: 2-Factor Authentication against Relay Attack on Passive Keyless Entry and Start System") disclose an ambient sound correlation and audible random sound emission scheme designed to compare acoustic environments between a vehicle and a key. In contrast to Choi et al., which relies upon audible acoustic emissions and ambient background noise sampling - mechanisms highly susceptible to environmental acoustic noise and disruptive to vehicle occupants - the present invention utilizes an inaudible, near-ultrasonic Continuous-Phase Frequency-Shift Keying (CP-FSK) payload 230 that is steganographically embedded beneath a low-pass-filtered vehicle audio carrier 210 (such as a welcome chime) with an unmodulated spectral guard band 220. Furthermore, the present invention integrates directly with an ECU multi-stage state machine (Stage 1 to Stage 2 ignition gating), an unpowered dormant fallback mode 440 preventing parallel brute-force attacks, and a locally-encrypted cryptographic key vault architecture in which the plaintext master secret is never persisted to non-volatile storage (FIG. 5), elements entirely absent from ambient acoustic correlation schemes.

3.5 Differentiation from Ultrasonic Point-of-Sale Communication (US Patent 9,098,865 B2 - Malbec Labs)

[0018]
US 9,098,865 B2 describes near-field communication systems utilizing unmasked ultrasonic bursts between 16 kHz and 22 kHz for commercial point-of-sale (POS) financial transactions and data transfer in open retail spaces. In contrast to retail POS systems, the present invention is directed to automotive immobilization and anti-relay protection, utilizing the physical structural boundary 100 of an enclosed vehicle cabin to trap acoustic signals. The present invention does not transmit unmasked, raw ultrasonic bursts in open air; rather, it executes steganographic spectral masking beneath a low-pass-filtered audible vehicle carrier 210, creates a dedicated spectral guard band 220, interfaces with vehicle CAN/LIN bus immobilizer interlocks 135, 140, enforces sub-second latency constraints (FIG. 6), and provides role-based hidden technician override gateways (460, 470).

3.6 Differentiation from In-Cabin Ultrasonic Positioning Prior Art (US Patent Application 2019/0025402 A1)

[0019]
US 2019/0025402 A1 describes systems utilizing ultrasonic acoustic signals (16 kHz to 26 kHz) to determine the spatial location or zone of a mobile device within a vehicle cabin (e.g., driver seat vs. passenger seat) via acoustic ranging. In contrast to in-cabin acoustic localization and distance-ranging systems, the present invention operates independently of device localization, spatial triangulation, or acoustic distance measurement. The present invention utilizes in-cabin acoustic transmissions strictly as an out-of-band cryptographic authentication transport layer to gate multi-stage ECU immobilizer circuits 140.

3.7 Differentiation from General Audio Token Prior Art (US Patent 7,533,735 B2 and US Patent 7,966,497 B2 - Qualcomm Inc.)

[0020]
US 7,533,735 B2 and US 7,966,497 B2 describe standalone acoustic challenge-response tokens in the audible range (1 to 3 kHz) for general data access and financial transactions. The present invention distinguishes itself by providing an out-of-band steganographic concealment system, wherein near-ultrasonic Continuous-Phase Frequency-Shift Keying (CP-FSK) payloads 230 are spectrally masked beneath an audible, low-pass-filtered vehicle audio carrier 210 (such as a welcome chime) to interface directly with multi-stage automotive ECU immobilizer logic (FIG. 4).

SUMMARY OF THE INVENTION

[0021]

The present invention provides a physical-presence authentication method and architecture that utilizes re-existing in-cabin audio hardware to establish an anti-relay and anti-theft firewall.

[0022]
The present invention operates effectively as a standalone physical-presence immobilizer, and seamlessly establishes an out-of-band Two-Factor Authentication (2FA) security layer that complements primary wireless mechanisms (such as standard Low Frequency/Ultra High Frequency RF fobs, Bluetooth Low Energy digital keys, Near-Field Communication, or Ultra-Wideband systems 150).

[0023]
In two-factor embodiments, primary access mechanisms (RF, BLE, UWB, or NFC 150) serve as Factor 1, authorizing initial vehicle door unlock/ingress and triggering an ECU wake-up state (Stage 1, 410). The present system provides Factor 2, requiring an in-cabin acoustic token emitted by a secondary authorized device 110 (such as a user smartphone, smartwatch, or wearable) to authorize vehicle propulsion and ignition systems (Stage 2, 430).

[0024]
Dual-Threat Elimination:

[0025]
Mitigating RF Relay Attacks: External wireless amplifier bridges 160 fail because the acoustic secondary token is physically contained and attenuated by the vehicle cabin boundary 100, preventing an external receiver from capturing a usable signal-to-noise ratio.
[0026] Mitigating Physical Key Theft: Even if an unauthorized party physically steals the authentic primary key fob 150, vehicle engine ignition is prevented because the ECU 130 is configured to inhibit propulsion circuits 140 unless the secondary in-cabin acoustic credential is cryptographically validated.

[0027]
The system comprises the following inventive principles:

[0028]
Acoustic Boundary Attenuation & Containment: Near-ultrasonic acoustic frequencies (16.0 kHz to 22.0 kHz) exhibit rapid physical air attenuation and suffer substantial transmission loss through laminated automotive safety glass, steel body panels, and elastomeric weatherstripping 100. An acoustic token emitted inside an enclosed vehicle cabin is attenuated below an operable signal-to-noise ratio (SNR) externally, such that an external receiver cannot obtain a reliably de-modulable authentication bitstream under the intended operating conditions.

[0029]
Steganographic Spectral Masking & Guard Band Isolation: The authentication token is modulated using Continuous-Phase Frequency-Shift Keying (CP-FSK 230) and layered beneath an audible acoustic carrier 210 (such as a multi-tone sinusoidal welcome chime, alert tone, or sound signature). The audible carrier is digital low-pass filtered (rolloff 215, cutoff frequency <= 14.0 kHz) to establish a dedicated, silent spectral guard band 220 (>= 1.5 kHz wide, situated between 14.0 kHz and 16.0 kHz), preventing audible acoustic harmonics from bleeding into the near-ultrasonic receiver detection bins.

[0030]
State-Gated Multi-Stage Ignition Authorization: The vehicle's ECU 130 implements a multi-stage security sequence:

[0031]
Locked State (400): High-voltage contactors, fuel pump relays, or starter motor solenoids 140 are physically isolated or software-inhibited.

[0032]
Stage 1 (Physical-Presence Listening State 410): Detection of a preliminary access or wake-up event (e.g., door handle capacitive touch, door opening, seat sensor, power-on button, or RF fob detection 402) wakes the ECU 130 and powers on an interior cabin microphone 120 in a listening state for an ingress time window (e.g., configurable between 10.0 and 60.0 seconds, and preferably 30.0 to 45.0 seconds).

[0033]
User Retry Capability (424): If an initial audio playback fails or is missed, pressing a vehicle start button executes an immediate retry trigger 424, refreshing or re-energizing the Stage 1 listening state 410 without locking out the driver.

[0034]
Stage 2 (Starter Authorization 430): The starter circuit 140 is unlocked only upon capture, demodulation, integrity verification, and cryptographic validation of the in-cabin acoustic token.

[0035]
Anti-Streaming and Bidirectional Challenge-Response Protection: To prevent remote acoustic streaming attacks (wherein an adversary attempts to bridge audio between a remote mobile device and a speaker placed in the cabin over a data network), the system provides sub-second time-epoch synchronization and an optional bidirectional acoustic challenge-response sequence (FIG. 6). In the bidirectional embodiment, the vehicle cabin speaker 125 emits a randomized acoustic challenge

seed inside the enclosed cabin that must be captured by the user device 110 to compute the valid response token within a strict round-trip latency window (Delta-t <= 1.5 seconds)

[0036]
State-Gated Dormant Fallback Override: To prevent user lockouts resulting from a depleted user device battery or local acoustic interference, a manual fallback interface 440 (e.g., steering-wheel button entry or infotainment PIN pad/password entry) is maintained. To eliminate parallel brute-force attack vectors, the manual fallback listener remains completely dormant and unpowered until the primary acoustic listening window reaches an explicit timeout condition/ failed attempts (426).

[0037]
Role-Based Diagnostic Gateway & 3-Empty-Entry Service Trigger: For maintenance personnel, emergency recovery operators, and fleet technicians, an authenticated diagnostic mode 470 provides access to forensic telemetry and overrides immobilizer locks. In a preferred embodiment, submitting an empty password entry three consecutive times (460) on the fallback interface reveals the technician login prompt. Access remains separately gated behind a cryptographic seed-key challenge-response sequence (ISO 14229 UDS 0x27), ensuring elevated privileges cannot be obtained without an authorized service credential.

[0038]
Zero-Additional-Transceiver Implementation: The system is executed via software and firmware on vehicle compute modules (ECU 130, Body Control Module, or Infotainment Head Unit), utilizing factory cabin hands-free microphones 120 and factory cabin audio speakers 125 present in modern automotive platforms.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039]
FIG. 1 is a block diagram illustrating the overall system architecture, showing the interaction between the user transmitting device 110, the enclosed vehicle acoustic boundary 100, the in-cabin microphone 120, the vehicle bus 135, and the vehicle Engine Control Unit (ECU / BCM 130).

[0040] FIG. 2 is a frequency spectrum diagram illustrating the steganographic spectral split, showing the low-pass-filtered audible carrier 210 (<= 14.0 kHz), the unmodulated spectral guard band 220 (14.0 to 16.0 kHz), and the near-ultrasonic CP-FSK data carrier peaks 232, 234 in the payload band 230 (16.0 to 22.0 kHz).

[0041]

FIG. 3 is a schematic diagram of the transmission packet framing structure 300, illustrating the alternating clock synchronization preamble 302, the unique Start-of-Frame (SOF) delimiter 304, the cryptographic payload 306, and the integrity checksum 308.

[0042]
FIG. 4 is a state-transition flowchart illustrating the ECU multi-stage ignition sequence, start-button user retries 424, rolling lockout timers 450, dormant fallback override 440, and the 3-empty-entry service trigger 460.

[0043]
FIG. 5 is a cryptographic architecture diagram illustrating a locally-encrypted key vault storage model utilizing PBKDF2 key derivation and AES-256 vehicle secret encapsulation on user device 500 and HSM 522 on ECU 520.

[0044]
FIG. 6 is a sequence diagram illustrating the bidirectional in-cabin acoustic challenge-response handshake between the vehicle audio system 602, the passenger cabin acoustic boundary 604, and the user mobile device 606.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
6.1 System Architecture & Hardware Integration

[0045]
Referring to FIG. 1, the system comprises an In-Cabin Transmitting Device 110 (e.g., a user smartphone, smartwatch, key fob, or smart wearable having acoustic transmitter 112) and a Vehicle Receiving Subsystem.

[0046]
The Vehicle Receiving Subsystem interfaces with:

[0047]
One or more pre-existing interior cabin microphones 120 (such as a hands-free telematics microphone, active noise cancellation headliner microphone, or dedicated cabin acoustic sensor).

[0048]
The vehicle infotainment head unit and interior acoustic transducers (cabin speakers 125).

[0049]
An Electronic Control Unit (ECU / BCM 130) or dedicated security module communicatively coupled across the vehicle CAN, CAN-FD, or LIN bus 135.

[0050]
In one embodiment, cryptographic keys, vehicle secrets 524, and session states are maintained within software-defined persistent stores. In a preferred embodiment, cryptographic operations, vehicle secrets 524, and session states reside within an automotive Hardware Security Module (HSM 132, 522) or Secure Hardware Extension (SHE) conforming to EVITA and ISO/SAE 21434 standards, operatively integrated with an ISO 14229-1 (UDS) diagnostic stack.

[0051]
The vehicle structure (comprising glass window assemblies, sheet metal panels, rubber door weatherstripping, and acoustic headlining) establishes a closed physical attenuation perimeter 100. Near-ultrasonic acoustic transmissions generated within the cabin undergo substantial structural transmission loss, attenuating signals below the ambient noise floor in the external threat space 160 outside the vehicle.

6.1.1 Powertrain Interlock Architectures & Secondary Immobilization Targets

[0051A]
The vehicle propulsion interlock 140 is operatively coupled to one or more physical and electronic vehicle subsystem actuators across diverse powertrain architectures, including Internal Combustion Engine (ICE) vehicles, Battery Electric Vehicles (BEVs), Plug-in Hybrid Electric Vehicles (PHEVs), and Fuel Cell Electric Vehicles (FCEVs).
In various embodiments, enabling or disabling the vehicle propulsion interlock comprises selectively gating one or more of the following immobilization targets:

1.
Electric Vehicle (EV/PHEV) High-Voltage Interlocks: Selectively isolating the high-voltage (HV) traction battery pack by commanding High-Voltage Main Contactors (positive and negative DC contactors, pre-charge relays, or pyro-fuses), inhibiting the low-voltage gate drivers of the high-voltage traction inverter (disabling the pulse-width modulation PWM switching signals to IGBT or SiC MOSFET power stages), or disabling the high-voltage DC-DC converter.

2.
Internal Combustion Engine (ICE) Starter & Fuel Gating: De-energizing the starter motor solenoid relay, physically disconnecting primary power to the in-tank or high-pressure direct-injection fuel pump via an electromechanical or solid-state fuel pump relay, or commanding the Engine Control Module (ECM) to software-inhibit spark ignition coil triggers and fuel injector solenoid pulses.

3.
Drivetrain & Shift-by-Wire (SbW) Mechanical Locks: Inhibiting the electronic transmission control unit (TCU) from shifting out of Park ("Park-Lock"), overriding electronic shift-by-wire rotary/stalk selectors, commanding an Electronic Steering Column Lock (ESCL) motorized deadbolt to remain engaged, or commanding the Electronic Parking Brake (EPB) control module to maintain full caliper clamping pressure regardless of accelerator pedal input.

4.
Drive-by-Wire & Throttle Signal Clamping: Intercepting and clamping Accelerator Pedal Position (APP) sensor voltage signals to a neutral/zero-torque state via an electronic analog switch or gateway filter, preventing drive torque generation even if the powertrain remains powered.

5.
In-Vehicle Network & Gateway Isolation: Utilizing the Central Gateway Module (CGM) or Body Control Module (BCM) to suppress or drop authorized propulsion-enablement CAN/LIN messages (such as AUTOSAR Secure On-Board Communication / SecOC cryptographic message authentication codes) directed to the drive inverter, engine management system, or power distribution box e-fuses/smart high-side power switches (e.g., PROFET devices).

6.2 Digital Signal Processing (DSP) & Steganographic Modulation

[0052]
To eliminate phase discontinuities that cause audible transient clicks and wideband spectral splatter, the transmission payload 230 is modulated using continuous-phase binary frequency-shift keying (CP-BFSK), hereinafter referred to consistently as CP-FSK.

The instantaneous phase θ(t) is accumulated continuously across symbol transitions:

$\theta(t) = 2\pi \int_0^t f(\tau)\, d\tau$
(instantaneous phase theta(t) equals two times pi times the integral from zero to t of frequency f(tau) d-tau)

[0053]
The carrier frequencies f0 (binary '0', 232) and f1 (binary '1', 234) are selected from near-ultrasonic frequency pairs situated between 16.0 kHz and 22.0 kHz, including:

[0054]
Pair A:
f0 = 16,000 Hz
f1 = 17,000 Hz
f0 equals 16,000 Hz, f1 equals 17,000 Hz
(broad commodity hardware compatibility)

[0055]
Pair B:
f0 = 18,000 Hz
f1 = 19,000 Hz
f0 equals 18,000 Hz, f1 equals 19,000 Hz
(near-inaudible high-isolation band)

[0056]
Pair C:
f0 = 18,500 Hz
f1 = 19,500 Hz
f0 equals 18,500 Hz, f1 equals 19,500 Hz
(stealth band)

[0057]
Pair D:
f0 = 20,000 Hz
f1 = 21,500 Hz
f0 equals 20,000 Hz, f1 equals 21,500 Hz
(ultrasonic band for 48 kHz / 96 kHz DACs)

[0058]
The bit duration Tb is configured between 5.0 ms and 20.0 ms (preferred: 10.0 ms, ≈100 baud).

[0059]
Each bit pulse is shaped using a Hann window function w[n] to suppress out-of-band spectral sidelobes:

$w[n] = 0.5 - 0.5 \cos(2\pi n / (Ns - 1))$, for $0 \leq n < Ns$
(w[n] equals 0.5 minus 0.5 times cos(2 times pi times n divided by (Ns minus 1)), for n from 0 up to Ns minus 1; where frame sample count Ns equals sampling rate fs multiplied by bit duration Tb)

[0060]
Ns represents the number of audio samples per bit period:

$Ns = fs \cdot Tb$
(frame sample count Ns equals audio sampling rate fs multiplied by bit duration Tb)

[0061]
where fs is the audio sampling rate (44,100 Hz or 48,000 Hz).

[0062]
The audible masking layer 210 is sampled at fs = 44,100 Hz or 48,000 Hz.

[0063]
The audible carrier comprises a multi-tone sinusoidal chime (1–5 seconds, e.g., 4 seconds). Using tones < 4,000 Hz minimizes harmonic distortion and keeps the 14–16 kHz guard band clean.

[0064]
The audible layer is processed through a digital low-pass filter:

[0065]
Filter topology: 12th-order Butterworth LPF (six SOS biquads)

[0066]
Cutoff frequency fc: 12,000–14,500 Hz (preferred: 14,000 Hz)

[0067]
Stopband attenuation: >80 dB at 16,000 Hz, establishing a ≥1.5 kHz guard band below the ultrasonic carriers.

[0068]
Amplitude scaling:
audible peak ≈ 0.65
ultrasonic peak ≈ 0.25
Composite peak ≤ 0.90

$\max(|x_{audible}[n] + x_{ultrasonic}[n]|) \leq 0.90$
(maximum of audible signal plus ultrasonic signal less than or equal to 0.90 of full scale)

6.3 Packet Framing Structure and Protocol Format

[0069]
The transmitted bitstream comprises:

[0070]
Clock Synchronization Preamble (32 bits): 10101010...

[0071]
Start-of-Frame (SOF) Delimiter (16 bits): 1110001110001110

[0072]
Cryptographic Payload: rolling nonce + HMAC-SHA256 (256-bit key)

[0073]
Integrity Checksum: CRC32 (IEEE 802.3, polynomial 0xEDB88320)

6.4 Demodulation via Non-Coherent Quadrature (I/Q) Detection

[0074]
Non-coherent I/Q detection eliminates phase-cancellation issues.

[0075]
The received signal is band-pass filtered (15.5–22.5 kHz).

[0076]
For each bit window Ns:

$I0 = \Sigma\, x[n] \cdot \sin(2\pi\, f0\, n / fs)$
$Q0 = \Sigma\, x[n] \cdot \cos(2\pi\, f0\, n / fs)$

$I1 = \Sigma\, x[n] \cdot \sin(2\pi\, f1\, n / fs)$
$Q1 = \Sigma\, x[n] \cdot \cos(2\pi\, f1\, n / fs)$

(In-phase component I0 equals summation of sample x[n] times sin(2 pi f0 n / fs); Magnitude M0 equals I0 squared plus Q0 squared. Decoded bit equals 1 if magnitude M1 is greater than M0, otherwise decoded bit equals 0)

[0080]
Energy magnitudes:

$M0 = I0^2 + Q0^2$
$M1 = I1^2 + Q1^2$

[0082]
Bit decision rule:

DecodedBit = '1' if M1 > M0
DecodedBit = '0' if M0 ≥ M1

[0083]
This ensures robust bit recovery regardless of phase angle.

6.5 Multi-Microphone Diversity Combining

[0084]
When multiple cabin microphones exist, the ECU performs diversity combining.

[0085]
Selection Diversity:
ECU measures SNR during preamble and selects the microphone with highest SNR.

[0086]
Maximal Ratio Combining (MRC):
M0 and M1 are computed per microphone, weighted by channel SNR, and summed before bit decision. This mitigates acoustic nulls caused by passenger positioning or cabin obstructions.

6.6 Multi-Stage Automotive ECU State Machine & Dual-Factor Gating

[0087]
Referring to FIG. 4, the ECU immobilizer operates according to a state-gated sequence enforcing two-factor authentication:

[0088]
State: Locked (Immobilizer Active, 400): Propulsion systems, fuel pump relays, starter motor circuits, and High-Voltage (HV) traction contactors 140 are physically isolated or software-disabled.

[0089]
Stage 1 (Physical-Presence Listening State, 410): Detection of a preliminary access or wake-up event (e.g., door handle touch, door opening, seat occupancy sensor, start button press, or RF key presence 402) transitions the ECU 130 to Stage 1. Interior cabin microphones 120 are energized and placed into an acoustic listening state. An acoustic ingress timer (e.g., configurable between 10.0 and 60.0 seconds, and preferably 30.0 to 45.0 seconds) is initiated to allow sufficient user interaction time to transmit the 4.0-second acoustic token.

[0090]
Token Validation and Decision Gating (420, 422): Audio captured by the interior microphone 120 during Stage 1 is evaluated by the ECU at a token validation decision point (420). If the acoustic token is successfully demodulated and cryptographically verified, the ECU proceeds via the successful-validation branch (YES, 422) to transition to Stage 2 (430). If token validation fails (NO), the failure is registered as a discrete unsuccessful attempt (e.g., Fail Attempt 1, Fail Attempt 2) without immediately exiting the Stage 1 listening state (410).

[0091]
User Retry Loop (424): If the user misses the window or fails an initial token transmission, pressing the vehicle engine Start button acts as an immediate retry trigger 424, refreshing or re-energizing the Stage 1 listening state 410 without locking out the user. Unsuccessful validation attempts occurring across successive invocations of the User Retry Loop (424) are counted by the ECU and, together with the state described in Section 6.7, inform the transition into Lockout Mode (450) upon a third consecutive incorrect attempt.

[0092]
Stage 2 (Starter Authorization, 430): Upon microphone capture, band-pass filtering across the active carrier band (15.5 to 22.5 kHz), SOF detection 304, CRC32 verification 308, and cryptographic validation of the token 306, the ECU transitions to Stage 2 (430). The starter motor relay or high-voltage contactor interlock 140 is enabled, permitting vehicle propulsion.

[0093]
In two-factor embodiments, the ECU 130 is explicitly configured such that possession of the primary RF/BLE key fob 150 alone is insufficient to transition the ECU from Stage 1 (410) to Stage 2 (430). If an authentic physical key fob is stolen and brought into the vehicle without the paired secondary acoustic device 110, the acoustic listening timer expires (426) and the ECU transitions to Fallback Mode 440, keeping the engine starter interlock 140 fully immobilized.

6.7 Anti-Tamper Lockout & State-Gated Fallback Modes

[0094]
If the acoustic listening window expires (426) without receiving a valid token (e.g., resulting from a depleted user mobile phone battery), the ECU enters Fallback Mode 440.

[0095]
Crucially, during normal acoustic listening (Stage 1, 410), all secondary manual PIN listeners (e.g., steering-wheel button decoders or center-console touchscreen interfaces) remain completely unpowered and dormant. This state-gated isolation eliminates parallel brute-force attack vectors during normal operation. The fallback interface awakens only after an explicit acoustic timeout event 426.

[0096]
The ECU tracks failed authentication attempts. If three (3) consecutive incorrect user PIN attempts occur (444) within a rolling 5-minute window:

[0097]
The ECU enters Lockout Mode 450, suspending all authentication processing for a 5-minute cooldown period.

[0098]
A security violation event is written to non-volatile memory alongside geographic telemetry coordinates (when telematics location services are enabled).

6.8 Locally-Encrypted Key Vault

[0099]
Referring to FIG. 5, persistent cryptographic credential storage on client devices 500 operates such that the plaintext master secret is never persisted, as follows:

[0100]
The user’s master secret or password 502 is never written to a non-volatile disk in plaintext.

[0101]
Key Derivation: The user password 502 and a 16-byte random cryptographic salt 504 are processed via PBKDF2 (HMAC-SHA256, minimum 100,000 iterations) to derive a 256-bit AES encryption key 506.

[0102]
The derived key 506 encrypts a permanent 256-bit vehicle secret 510 via AES-256-CBC with Initialization Vector (IV 508).

[0103]
Only the salt 504, IV 508, and AES ciphertext are retained in local persistent vault storage 512.

[0104]
Extraction of client storage data yields zero actionable plaintext keys without knowledge of the user password.

6.9 Event Logging, Telematics & Forensic Traceability

[0105]
Upon every authentication transaction (successful unlock, rejected token, service override, or lockout trigger), the ECU 130 records an immutable event entry in a non-volatile flash memory. The log entry contains:

[0106]
High-resolution real-time timestamp.

[0107]
Transaction classification (UserUnlock, FailedAttempt, ServiceMode, LockoutTriggered).

[0108]
Current ECU State (Locked, Stage1, Stage2, MaintenanceMode).

[0109]
Geographic position data (approximate GPS Latitude and Longitude obtained via an onboard Telematics Control Unit, where available and privacy-authorized).

[0110]
These records enable forensic verification following attempted vehicle theft or unauthorized service access.

6.10 ECU Pairing & Unique Vehicle Binding

[0111]
During vehicle manufacturing, dealer commissioning, or owner pairing:

[0112] A unique Vehicle Identification Number (VIN 526) and a cryptographically randomized 256-bit Vehicle Secret 524 are provisioned within the ECU's Secure Hardware Extension (SHE) or Hardware Security Module (HSM 132, 522).

[0113]

A pairing certificate 528 is constructed via an HMAC-SHA256 hash:
Pairing Certificate 528 = HMAC-SHA256 (Vehicle Secret 524,VIN 526)

[0114]

Acoustic tokens generated by authorized user devices 110 incorporate this vehicle-specific cryptographic binding, preventing replay or reuse of a valid acoustic token across different vehicles.

6.11 Authenticated Service / Emergency Retrieval Mode & 3-Empty-Entry Trigger

[0115]

To accommodate vehicle maintenance, emergency recovery, and dealership diagnostics without exposing the owner's private credentials:

[0116]

Empty-Submission Technician Prompt Trigger (460): In a preferred embodiment, when in manual fallback mode 440, submitting a null-length or empty PIN entry three (3) consecutive times (460) automatically unmasks and reveals a hidden manufacturer service-center login terminal on the vehicle touchscreen display or diagnostic interface. Crucially, submitting three consecutive empty entries does not grant vehicle access, does not authorize engine starting, and does not disable vehicle immobilizer circuits on its own; rather, it functions solely as a hidden gateway trigger to unmask a restricted technician password entry screen. Said empty-submission trigger is treated by the ECU 130 as distinct from incorrect user PIN entries (444) and does not count toward user anti-brute-force lockouts. Alternatively, a pre-configured multi-modal physical switch sequence (such as depressing and holding the vehicle brake pedal switch continuously while cycling the hazard light switch five consecutive times and the engine start/stop button five consecutive times within an 8.0-second temporal window) may be utilized to unmask said technician terminal.

[0117]

Manufacturer Service Center Authentication Terminal (470): Once revealed, the technician terminal requires the manual entry of an authorized manufacturer master service password, a dealership emergency retrieval PIN, or the execution of an authenticated diagnostic challenge conforming to ISO 14229-1 (Unified Diagnostic Services Service 0x27 SecurityAccess). The ECU generates a randomized

challenge seed for an attached service tool or prompts for the manufacturer's secure facility code.

[0118]
Elevated Privileges & Starter Authorization: Vehicle starting (Stage 2, 430) and propulsion circuits 140 are unlocked if and only if the technician enters a valid manufacturer service-center password/PIN or returns an authorized cryptographic response computed via a master OEM service key. Upon successful validation of the manufacturer credential, the ECU transitions to Service / Emergency Retrieval Mode 470, authorizes temporary vehicle starting (Stage 2, 430), clears anti-tamper lockouts 450, and grants read-only access to forensic telemetry logs while keeping the owner's personal encrypted master key strictly protected.

6.12 Bidirectional In-Cabin Acoustic Challenge-Response & Latency Gating

[0119]
Referring to FIG. 6, in a preferred anti-streaming embodiment:

[0120]
Upon entering Stage 1, the vehicle infotainment audio system 602 emits an acoustic challenge chirp comprising a randomized cryptographic challenge seed into the passenger compartment 604 via the cabin speakers 125. The challenge seed is emitted either as an audible acoustic chirp or as a near-ultrasonic acoustic burst, and includes an incremental cryptographic freshness counter.

[0121]
The user device microphone 606 captures this in-cabin challenge seed.

[0122]
The user device 606 computes an acoustic response token cryptographically bound to said challenge seed, the freshness counter, and the vehicle secret 510.

[0123]
The user device 606 transmits the response token near-ultrasonically into the cabin 604.

[0124]
The ECU microphone 120 captures the acoustic token. The ECU hardware timer calculates the round-trip latency Δt from the completion of the challenge chirp emission to the receipt of the response token.

The ECU verifies that the latency threshold is satisfied:

$\Delta t \leq 1.5$ seconds (round-trip latency delta-t less than or equal to 1.5 seconds)

[0125]
Upon validating both the cryptographic token and confirming that $\Delta t \leq 1.5$ seconds, the ECU transitions to Stage 2, enabling the propulsion interlock 140. (round-trip latency delta-t less than or equal to 1.5 seconds)

[0126]
Because an adversary streaming audio across cellular networks, VoIP bridges, or Internet connections incurs codec compression (which filters out frequencies above 14.0 kHz) and network transport latency typically exceeding 300 ms to 1000 ms, remote audio streaming relays fail both the latency threshold and cryptographic freshness tests without requiring external NTP clock synchronization.

6.13 Automotive Architecture Mapping (ISO 14229 / UDS / AUTOSAR)

Secure Key Storage (132, 522)	**Hardware Security Module (HSM) / Secure Element**	**EVITA / ISO/SAE 21434**
Stage 1 Session Request (410)	**Diagnostic Session Control (0x10)**	**ISO 14229-1 (UDS)**
Stage 2 Verification (430)	**SecurityAccess Seed/Key (0x27)**	**ISO 14229-1 (UDS)**
Acoustic Token Channel (230)	**Out-of-Band Physical Secondary Layer**	**Near-Ultrasonic Steganography**
Locked State (140, 400)	**Engine Immobilizer Relay / High-Voltage Interlock**	**ISO 26262 (ASIL-B/D)**
Service Override Gateway (470)	**Extended Diagnostic Security Session**	**ISO 14229-1 Service 0x27**

[0127]
Advanced Modulation Schemes: In high-speed alternative embodiments, Orthogonal Frequency-Division Multiplexing (OFDM), Chirp-FSK (Linear Frequency Modulation), or 4-FSK may be utilized across the 16.0 kHz to 22.0 kHz band 230.

[0128]
Wearable Transmitting Devices: The transmitting acoustic transducer 112 may be incorporated into a smartwatch, fitness band, key fob pendant, smart ring, or smart jewelry.

[0129]
Commercial, Aviation & Marine Applications: The system and architecture may be deployed within commercial haulage vehicles, agricultural tractors, earthmoving and construction equipment, light aircraft cockpits, commercial aviation flight decks, and marine yachts to provide physical presence verification and prevent unauthorized machine startup.

CLAIMS

What is claimed is:

1. A method for vehicle immobilization and relay-attack mitigation, comprising:

(a) detecting, by an electronic control unit of a vehicle, a preliminary vehicle access or wake-up event, and transitioning the electronic control unit into a physical-presence listening state in which at least one interior cabin microphone is energized for a predetermined ingress listening window while maintaining a vehicle propulsion interlock disabled;

(b) receiving, via said at least one interior cabin microphone, an acoustic authentication token emitted by a transmitting device within an enclosed passenger cabin of the vehicle, the acoustic authentication token comprising a near-ultrasonic Continuous-Phase Frequency-Shift Keying payload modulated across near-ultrasonic frequencies, wherein said payload is steganographically masked beneath an audible, low-pass-filtered vehicle audio carrier and separated from said carrier by an unmodulated spectral guard band;

(c) demodulating the Continuous-Phase Frequency-Shift Keying payload using non-coherent In-Phase and Quadrature energy detection across discrete near-ultrasonic frequency bins;

(d) verifying transmission integrity of the demodulated payload by validating a predefined Start-of-Frame delimiter and evaluating a cyclic redundancy checksum;

(e) cryptographically validating the demodulated payload against a vehicle-specific secret stored within a secure hardware module of the vehicle; and

(f) transitioning the electronic control unit to an authorized ignition state enabling said vehicle propulsion interlock only upon successful completion of steps (a) through (e), wherein the acoustic authentication token is physically contained within the enclosed cabin boundary such that external acoustic propagation of the token undergoes structural transmission loss that attenuates the token below an operable signal-to-noise ratio for external demodulation, the method operating independently of acoustic time-of-flight distance calculations.

2. The method of claim 1, wherein the predetermined ingress listening window is between 30.0 seconds and 45.0 seconds, the near-ultrasonic frequencies are

between 16.0 kHz and 22.0 kHz, and the unmodulated spectral guard band is at least 1.5 kHz wide.

3. The method of claim 1, wherein said audible, low-pass-filtered vehicle audio carrier is processed through a digital low-pass filter having a cutoff frequency less than or equal to 14.0 kHz to establish said unmodulated spectral guard band between 14.0 kHz and 16.0 kHz, wherein said digital low-pass filter comprises a 12th-order Butterworth filter implemented as a cascade of six second-order biquad sections providing greater than 80 dB attenuation at 16.0 kHz.

4. The method of claim 1, wherein carrier frequencies of said Continuous-Phase Frequency-Shift Keying payload are selected from the group consisting of: 16.0 kHz and 17.0 kHz; 18.0 kHz and 19.0 kHz; 18.5 kHz and 19.5 kHz; and 20.0 kHz and 21.5 kHz.

5. The method of claim 1, wherein each symbol of said Continuous-Phase Frequency-Shift Keying payload has a duration between 5.0 ms and 20.0 ms, corresponding to a nominal baud rate between 50 and 200 baud, and is pulse-shaped using a Hann window.

6. The method of claim 1, wherein said preliminary vehicle access or wake-up event comprises detection of a primary wireless key comprising at least one of a radio-frequency key fob, a Bluetooth Low Energy digital key, a Near-Field Communication key, or an Ultra-Wideband transceiver, and wherein said acoustic authentication token operates as an out-of-band second-factor authentication layer.

7. The method of claim 6, wherein the electronic control unit is configured to inhibit enablement of said vehicle propulsion interlock when said primary wireless key is detected in the absence of valid cryptographic verification of said acoustic authentication token.

8. The method of claim 1, further comprising:
Initializing an acoustic listening timer upon entering said physical-presence listening state; and transitioning said electronic control unit to a dormant fallback mode only upon expiration of said acoustic listening timer without receipt of a valid token, wherein said dormant fallback mode activates a manual in-cabin physical override interface that remains unpowered and unmonitored during said physical-presence listening state.

9. The method of claim 8, wherein pressing a vehicle engine start button prior to or during token playback executes a user retry loop that re-energizes said physical-presence listening state and refreshes said acoustic listening timer without locking out the user.

10. The method of claim 1, further comprising recording a timestamped audit log in non-volatile memory upon each authentication attempt, said audit log comprising an authentication status code and geographic telemetry data derived from an onboard telematics control unit.

11. The method of claim 1, wherein said vehicle-specific secret is provisioned within a Hardware Security Module or Secure Hardware Extension conforming to ISO/SAE 21434 and is cryptographically bound to a Vehicle Identification Number.

12. The method of claim 1, further comprising providing an authenticated workshop service mode conforming to ISO 14229 Unified Diagnostic Services, wherein receipt of an authorized cryptographic seed-key response or emergency retrieval PIN temporarily authorizes vehicle starting without requiring said acoustic authentication token, and wherein access to said authenticated workshop service mode is gated behind submitting an empty password entry three consecutive times on a manual fallback interface within the vehicle cabin.

13. The method of claim 1, wherein said Continuous-Phase Frequency-Shift Keying payload comprises a rolling cryptographic nonce bound to a discrete time epoch and an HMAC-SHA256 digest computed using said vehicle-specific secret.

14. The method of claim 1, wherein said electronic control unit disables said vehicle propulsion interlock for a predetermined cooldown interval following three consecutive incorrect authentication attempts within a rolling temporal window.

15. The method of claim 1, wherein said audible carrier comprises a multi-tone sinusoidal harmonic chime having a duration between 1.0 and 5.0 seconds, and wherein said Continuous-Phase Frequency-Shift Keying payload is mixed with said audible carrier at relative amplitudes scaled to constrain composite peak signal amplitude to less than or equal to 0.90 of digital-to-analog converter full scale.

16. The method of claim 1, wherein said vehicle comprises a plurality of interior cabin microphones, and wherein the electronic control unit executes spatial diversity combining across signals received from said plurality of interior cabin microphones by calculating non-coherent In-Phase and Quadrature energy magnitudes across each microphone channel and performing one of signal-to-noise selection diversity or maximal ratio combining prior to evaluating a bit decision rule.

17. The method of claim 1, wherein on a user transmitting device, a plaintext master password and a random salt are processed via PBKDF2 with a minimum of 100,000 iterations to derive an AES encryption key, said AES encryption key encrypting the

vehicle-specific secret for local persistent vault storage such that the plaintext secret is never written unencrypted to non-volatile memory.

18. A method for vehicle immobilization and relay-attack mitigation utilizing bidirectional acoustic challenge-response, comprising:

(a) detecting, by an electronic control unit of a vehicle, a preliminary vehicle access or wake-up event, and transitioning the electronic control unit into a physical-presence listening state while maintaining a vehicle propulsion interlock disabled;

(b) emitting, via an interior cabin speaker of the vehicle, an acoustic challenge seed comprising a randomized cryptographic value into an enclosed passenger cabin of the vehicle;

(c) receiving, via at least one interior cabin microphone, an acoustic response token emitted by a transmitting device situated within the cabin, wherein said acoustic response token is cryptographically derived from said acoustic challenge seed and modulated as a near-ultrasonic Continuous-Phase Frequency-Shift Keying payload steganographically masked beneath an audible carrier and separated by an unmodulated spectral guard band;

(d) measuring, via an internal timer of the electronic control unit, a round-trip latency from completion of emission of said acoustic challenge seed to receipt of said acoustic response token; and

(e) transitioning the electronic control unit to an authorized ignition state enabling said vehicle propulsion interlock only when said acoustic response token is cryptographically validated and said measured round-trip latency does not exceed a predetermined latency threshold,

wherein remote audio streaming relays over data networks are rejected upon exceeding said predetermined latency threshold.

19. The method of claim 18, wherein the near-ultrasonic frequencies are between 16.0 kHz and 22.0 kHz, the unmodulated spectral guard band is at least 1.5 kHz wide, the predetermined latency threshold does not exceed 1.5 seconds, and wherein said acoustic challenge seed further comprises an incremental cryptographic freshness counter cryptographically bound to said acoustic response token and a pre-shared 256-bit vehicle secret.

20. An automotive immobilizer system, comprising:

an electronic control unit operatively coupled across a vehicle communication bus to a vehicle propulsion interlock;

at least one interior cabin microphone communicatively coupled to said electronic control unit;
a secure hardware module operatively coupled to said electronic control unit and storing a 256-bit vehicle secret; and
a non-transitory computer-readable memory storing instructions that, when executed by said electronic control unit, cause said electronic control unit to:
(i) detect a preliminary vehicle access or wake-up event across said vehicle communication bus;
(ii) energize said at least one interior cabin microphone to enter a physical-presence listening state for a predetermined ingress listening window while maintaining said vehicle propulsion interlock disabled;
(iii) capture, via said at least one interior cabin microphone, an in-cabin acoustic authentication token comprising a near-ultrasonic payload modulated via Continuous-Phase Frequency-Shift Keying across near-ultrasonic frequencies and steganographically masked beneath an audible low-pass-filtered carrier separated by an unmodulated spectral guard band;
(iv) demodulate said near-ultrasonic payload via non-coherent quadrature energy detection across discrete near-ultrasonic frequency bins;
(v) verify transmission integrity of said payload using a Start-of-Frame delimiter and a cyclic redundancy checksum;
(vi) cryptographically validate said payload against said vehicle secret stored within said secure hardware module; and
(vii) enable said vehicle propulsion interlock only upon successful cryptographic validation of said payload, relying upon acoustic boundary attenuation and spatial confinement within a passenger cabin rather than acoustic distance calculations.

21. The system of claim 20, wherein the predetermined ingress listening window is between 30.0 seconds and 45.0 seconds, the near-ultrasonic frequencies are between 16.0 kHz and 22.0 kHz, and the unmodulated spectral guard band is at least 1.5 kHz wide.

22. The system of claim 20, wherein said preliminary vehicle access or wake-up event comprises detection of a primary wireless key, and wherein the electronic control unit is configured such that detection of said primary wireless key alone is insufficient to enable said vehicle propulsion interlock in the absence of valid cryptographic verification of said in-cabin acoustic authentication token.

23. The system of claim 20, wherein said secure hardware module comprises an automotive Hardware Security Module or Secure Hardware Extension conforming to ISO/SAE 21434, and wherein said at least one interior cabin microphone comprises a hands-free telematics microphone or an active noise cancellation headliner microphone.

24. The system of claim 20, wherein said electronic control unit is further configured to:

apply a digital band-pass filter encompassing active carrier frequencies between 15.5 kHz and 22.5 kHz to isolate said near-ultrasonic payload prior to demodulation;

enter a dormant fallback mode upon expiration of an acoustic listening timer, wherein manual fallback interfaces remain unpowered and software-isolated prior to said expiration; and

disable said vehicle propulsion interlock for a predetermined cooldown interval following three consecutive incorrect authentication attempts within a rolling temporal window.

25.

The system of claim 20, wherein said instructions further cause said electronic control unit to: enter a diagnostic maintenance mode conforming to ISO 14229 upon receipt of a valid cryptographic seed-key response or emergency retrieval PIN invoked via submitting an empty password entry three consecutive times on a fallback interface; and

cause a vehicle interior speaker to emit an acoustic challenge seed into the cabin upon entering said physical-presence listening state, and validate said acoustic authentication token only when received within a round-trip latency window not exceeding 1.5 seconds.

ABSTRACT

An out-of-band physical-presence authentication system and method for vehicle immobilization and relay attack mitigation. An electronic control unit (ECU 130) enters a physical-presence listening state upon detecting a wake-up event (402), energizing an interior cabin microphone (120) for an ingress listening window while maintaining propulsion interlocks (140) disabled.

An in-cabin device (110) emits an authentication token comprising a near-ultrasonic Continuous-Phase Frequency-Shift Keying (CP-FSK) payload (230) steganographically masked beneath a low-pass-filtered audible carrier (210) and separated by an unmodulated spectral guard band (220).

The ECU demodulates the token via non-coherent quadrature (I/Q) energy detection and cryptographically validates it against a vehicle secret (524) stored in a secure hardware module (132, 522) to enable propulsion (140). Physical security derives from acoustic boundary attenuation (100) within the enclosed cabin rather than distance ranging. Bidirectional challenge-response (FIG. 6), start-button retries (424), and dormant fallbacks (440) are also provided.

[Figure 4 to accompany the Abstract]